MIL-C-48259 (PA)
24 March 1975

MILITARY SPECIFICATION

CARTRIDGE, 40MM, PRACTICE, M407A2

LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Cartridge designated as Cartridge, 40MM, Practice, M407A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Method of
MIL-P-20444	- Primer, M42
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

8835104	- Box, Wirebound, Packing, Ammunition for Cartridge, 40MM
8835105	- Box, Fiber, Packing, Ammunition for Cartridge, 40MM
8844609	- Cartridge Case Loading Assembly, M118
9287439	- Cartridge, 40MM, Practice, M407A2
9287440	- Projectile Assembly
9287441	- Body and Skirt Loading Assembly
9287446	- Smoke Pellet

INSPECTION EQUIPMENT DRAWINGS

8827889	- Chamber Gage
8827890	- Alignment Check
8827895	- Limit checks
9202770	- Chamber Gage (Alternate)
9202783	- Alignment check (Alternate)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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TECP-700-700 Volume 4 - TECOM Material Test Procedure

CODE OF FEDERAL REGULATIONS

. Title 49 - Transportatation 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on Drawing (Dwg.) 9287439, all associated drawings, and with all requirements in applicable specifications.

3.3 Primer functioning.-The primer shall function without evidence of squibbing or hangfire. Paragraph 6.6 of MIL-P-20444B, Amendment 3 contains definitions of functioning failures.

3.4 Functioning.-

3.4.1 Cartridge.-The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 247 plus or minus 7 feet per second (fps) and a standard deviation not exceeding 4.5 fps.

3.4.2 Safety.-There shall be no premature burst (see 6.8) or evidence thereof in any of the test firing.

3.5 X-ray examination of ballistic samples.-Prior to forwarding the sample cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination.

3.6 X-ray of fuze assemblies prior to assembling to body and skirt assemblies.-The fuze assembly shall be x-rayed for improperly assembled, missing parts, armed or partially armed.

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3.7 Production control check on loaded M118 cartridge cases.- There shall be no evidence of a primer malfunction.

3.8 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.9 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract (see 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following type of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 First Article Inspection.-Not applicable to this specification.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078 (PA). Each inspection lot shall contain:

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a. Cartridge case metal parts from one interfix lot number from one manufacturer.
b. Body and skirt assemblies from one interfix lot number from one manufacturer.
c. Fuzes from one interfix lot number from one manufacturer.
d. Primers from one lot number from one manufacturer.
e. Propellant from one lot number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078 (PA).

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cartridge case loading assembly prior to assembling to projectile and prior to assembling base plug and primer				8844609
					NEXT HIGHER ASSEMBLY 9287439
CATEGORY	**EXAMINATION OR TEST**	**NO. OF SAMPLE UNITS**	**AQL OR 100%**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE /INSPECTION METHOD**
Critical					
1	Powder charge cup missing		100%	3.2	Gage
2	Spilled powder from cartridge case		100%	3.2	Visual
3	Propellant charge weight obviously incorrect (See 6.10)		100%	3.2	Visual
Special	None defined.				
Major A					
101	Propellant charge weight obviously incorrect (See 6.10)		100%	3.2	Visual
Major B	None defined				
Minor	None defined.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Cartridge case loading assembly, prior to assembling to projectile				8844609
					NEXT HIGHER ASSEMBLY 9287439
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer above flush with respect to base of case		100%	3.2	Gage
Special					
*a	Vent hole blocked or missing		100%	3.2	Automatic Gage or Visual/Equipment
Major A					
101	Base plug above flush		0.40%	3.2	Gage
102	Depth to base plug, max.		0.40%	3.2	Gage
103	Primer above flush or more than .005 below with respect to base plug		0.40%	3.2	Gage
104	Primer damaged		0.40%	3.2	Visual
Major B	None defined				
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

*Prior to case loading

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287446
4.4.2.3	Smoke Pellet				NEXT HIGHER ASSEMBLY 9287441
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Weight		0.40%	3.2	Balance
132	Height		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287441
4.4.2.4	Body and Skirt Loading Assembly, prior to assembling cup retaining				NEXT HIGHER ASSEMBLY 9287439
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B 131	Smoke pellet missing		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.5	Body and Skirt Loading Assembly		1	1	9287441
					NEXT HIGHER ASSEMBLY 9287439

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B 131	Depth to bottom of cup, retaining		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287440
4.4.2.6	Projectile Assembly				NEXT HIGHER ASSEMBLY 9287439
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A					
101	Weight incorrect		0.40%	3.2	Balance
102	Security of crimp of fuze		2.5%	3.2	4.5.5
Major B	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287439
4.4.2.7	Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A					
101	Chamber gage failure (after complete cure)		100%	3.2	8827889, 8827890, 8827895 / Alternate 9202770, 9202783
102	Security of projectile	80	1.50%*	3.2	4.5.6
103	Air pressure	200	1.50%	3.2	4.5.9
Major B	None defined				
Minor					
201	Total length		0.65%	3.2	Gage
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

*In addition, if any cartridge tested has a value of less than fifty (50) pounds or any cartridge case or projectile shows any evidence of uncured sealant after completion of this test, the lot shall be rejected.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Unsealed fiber box				8835105
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Improper number of cartridges		0.40%	3.2	Visual
132	Separator missing		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Sealed fiber box, prior to sealing bag	1 OF 1	8835105
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Heat seal test of packing seals	(a)	(a)	3.2	4.5.7
132	Container damaged		0.40%	3.2	Visual
133	Corners not blunted		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Marking misleading or unidentifable		0.65%	3.2	Visual

NOTES:

(a) Sampling and rejection shall be in accordance with MIL-P-116.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Unsealed wooden packing box				8835104
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Marking of bag misleading or unidentifiable		0.40%	3.2	Visual
132	Bag torn or perforated		0.40%	3.2	Visual
133	Bag improperly sealed		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Sealed wooden packing box				8835104
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Box damaged		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Visual
202	Car seal missing or improperly positioned		0.65%	3.2	Manual
203	Marking misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48259 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Cup, Retaining				9287447
					NEXT HIGHER ASSEMBLY 9287441
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	(See 4.5.1)			
Special	None defined				
Major A	None defined				
Major B					
131	Outside diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Height		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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4.4.3 Testing.-

4.4.3.1 Check test for deterioration of primers (See MIL-P-20444).-If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two (2) years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-P-20444. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into cartridge (See 6.6).

4.4.3.2 Holding fixture vent check.-The cartridge case holding fixture used in the base plug insertion operation shall have its air vents check per shift for any obstruction. If a vent is found to be closed or clogged, the vent shall be cleared and the previous hour's production shall be x-rayed 100 percent to ensure that the propellant charge cups have not been damaged. Cartridge cases found to have damaged charge cups shall be classed defective and removed from the lot.

4.4.3.3 Propellant weight of cartridge case.-The propellant weight shall be determined and then check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing or determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirement specified on Dwg. 8844609 shall be classified defective and removed from the lot.

4.4.3.4 Propellant weight of cartridge case, Alternate method.-At the start of production, three hundred eighty (380) consecutive samples per station, per machine shall be selected and weighed 100 percent. All samples shall comply with the applicable drawing requirement. All samples shall comply with the drawing requirement, five (5) samples, per station, per every two (2) hours production shall be selected and

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the average weight of the five (5) samples must meet the assessed mean propellant weight within ± 4 milligrams. If the average weight fails the requirement, correction will be made and three hundred eighty (380) consecutive samples weighed from that station. If all meet the requirement the sampling every two (2) hours will be resumed.

4.4.3.5 X-ray examination of fuze assemblies prior to assembling to body and skirt assembly (See Para. 3.6).-Each fuze assembly, prior to assembling to body and skirt assembly, shall be subjected to this test. If any fuze assembly shows evidence of being improperly assembled having missing parts, being armed or partially armed, it shall be classed defective and removed from the lot.

4.4.3.6 P roduction control check on loaded M118 Cartridge Case (See Para. 3.7).-Five (5) loaded M118 cases shall be randomly selected from every two (2) hours production from each primer assembly machine. The loaded M118 cases produced by each primer assembly machine shall be kept segregated and identified. They shall not be used until this test has been performed. If any sample fails to comply with the applicable requirement, the two hours production represented by the sample shall be rejected.

4.4.3.7 Security of crimp of base plug on cartridge case assembly (See Dwg. 8844609).-Twenty (20) each of cartridge case, base plug, and powder charge cup less propellant shall be selected from each lot and subjected to this test. If the crimped area of one or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected.

4.4.3.8 Primer functioning (See Para. 3.3).-Sampling shall be in accordance with Paragraph 4.2.3.2.2 of MIL-P-20444B, except that the sample size shall be eight hundred (800) primers from each lot. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected. Primers not originally purchased to this requirement shall be tested for the requirement of Paragraph 3.3.

4.4.3.9 Functioning (see paragraph 3.4).-Beginning with the first lot produced and continuing until three (3) consecutive lots have been accepted under the criteria specified in 4.4.3.9.1,

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the Government inspector shall select one hundred and forty-four (144) cartridges from each lot for this test. The cartridges shall be fired using three (3) randomly selected launchers, forty-eight (48) cartridges per launcher, to impact at approximately 200 meters from the launcher on six inch thick loose gravel pad.

4.4.3.9.1 Rejection.-The lot shall be rejected if:

a. Twelve (12) or more fail to function at initial impact. (No emission of smoke shall be classified as a low order failure and be placed in a dud category), (Major Defect).

b. Any primer blows back (as evidenced by perforation of the cup or extrusion of primer around firing pin causing difficulty in opening weapon), (Major defect).

c. Any premature burst occurs (see 6.8) (Critical defect)

d. Mean velocity and or standard deviation fail to meet the applicable requirements. (Major Defect).

e. Any primer misfires. (Major Defect)

f. (1) Thirty (30) or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor (Major Defect)

(2) One (1) or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor (Major Defect)

g. Any projectile sticks in gun bore or fails to attain 60 percent of the mean velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered a failure). (Major Defect).

h. Three (3) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major Defect).

i. Any metal part separation occurs prior to impact (Major Defect

j. Any premature burst occurs (critical defect).

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same

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approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is not significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

4.4.3.9.2 Suitability test plan (informational test only).- Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) consecutive lots that are accepted in accordance with Paragraph 4.4.3.9, an extra sample of ninety (90) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The ninety (90) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After the temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds + 165F ± 5 degrees F.
30 rounds ambient (between +40 and +110 degrees F).
30 rounds -65 degrees F ± 5 degrees F.

d. Fire the rounds in accordance with Paragraph 4.5.10 and with TECOM Material Test Procedures, TECP-700-700 Volume 4. Test results, failure analysis reports, and other information as requested shall be forwarded to Picatinny Arsenal, Attn: SARPA-QA-A-S for evaluation.

4.4.3.10 Functioning (See Paragraph 3.4) Regular Production.- After three (3) consecutive lots have met the criteria of 4.4.3.9.1, the Government inspector shall select seventy-two (72) cartridges from each lot for this test. The cartridges shall be fired using three (3) randomly selected launchers (twenty-four) cartridges per launcher) to impact at approximately 200 meters from the launcher and six inch thick loose gravel pad.

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4.4.3.10.1 Rejection.-The lot shall be rejected if:

a. Four (4) or more fail to function at initial impact (No emission of smoke shall be classified as a low order failure and be placed in a dud category,) (Major Defect)

b. Any primer blows back (as evidenced by perforation of the cup or extrusion of the primer around firing pin causing difficulty in opening weapon. (Major Defect)

c. Any premature burst occurs (see 6.8) (Critical defect)

d. Mean velocity and/or standard deviation fail to meet the applicable requirement (Major Defect)

e. Any primer misfires (Major Defect)

f. (1) Five (5) or more cartridges cases cannot be ejected from the launcher by spring operation of the extractor (Major Defect)

(2) One (1) or more cartridge cases cannot be extracted from the launcher by hand operation of the extractor (Major Defect)

g. Any projectile sticks in the gun bore or fails to attain 60 percent of the mean uncorrected velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered as a failure. (Major defect)

h. Three (3) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major defect).

i. Any metal part separation occurs prior to impact (Major Defect

j. Any premature burst occurs (Critical defect).

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

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4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.-

4.5.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Production control check on loaded M118 cartridge cases.-The loaded M118 cartridge case shall be fired from an approved launcher placed on an approved mount. Observation shall be made for primer functioning. This test is a destructive test. Parts so tested shall not be returned to the lot. NOTE: No projectile is required for this test.

4.5.3 X-ray examination of fuze assemblies prior to assembling to body and skirt assembly.-Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by x-ray equipment or any other method satisfactory to the Contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot.

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4.5.4 Security of crimp of base plug on cartridge case assembly.- The assembly shall be placed in an approved fixture and the minimum unseating force specified on dwg. 8844609 shall be applied to failure. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.5 Security of crimp of fuze.-The assembly shall be placed in an approved fixture and the minimum unseating force specified on dwg. 9287440 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.6 Security of projectile.-The cartridge shall be placed in an approved fixture and the axial force specified on dwg. 9287439 shall be applied. Projectile must separate from the cartridge case in the range specified on dwg. 9287439. This test is a non-destructive test. Parts so tested may be returned to the lot. Stake shall be removed from the loaded case prior to returning to the lot.

4.5.7 Heat seal test of packing seals.-This test shall be performed in accordance with MIL-P-116. Samples so tested shall not be returned to the lot.

4.5.8 Chamber gage failure.-The cartridge shall be placed in the chamber gage and must meet the requirement specified on dwg. 9287459. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.9 Air pressure.-The cartridge shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.10 Functioning.-This test shall be conducted at a Government owned proving ground. The rounds being tested shall be temperature conditioned at 70°F + 5°F for at least eight (8) hours, then fired within fifteen (15) minutes after being removed from the temperature soak. Gravel: The cartridge shall be fired to impact on a six inch thick loose gravel pad. The gravel pad used shall have a 1/2 inch maximum stone size.

The velocity shall be measured at twenty (20) feet from the muzzle of the launcher, using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 20 feet from the launcher and the other set 15 and

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25 feet from the launcher. Note: To insure that the chronographs are read properly. The ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more than plus or minus 0.005 the velocity results for that round shall be discarded and exluded from the calculation of mean velocity and velocity standard deviation (no more than four (4) rounds in a sample size of 72 or eight (8) rounds in a sample size of 144 may be disregarded). If the applicable number of improper ratios is exceeded additional rounds must be fired. Observation shall be made and results recorded for functioning mean velocity and standard deviation. The average of all rounds fired per sample (except above noted improper ratio(s)) shall meet the requirements the requirements of 3.4.1 (see 6.7).

4.5.10.1 Test Validity.-If, for any reason, proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.5.11 Primer functioning.-Each primer of the 800 samples shall be tested as specified in MIL-P-20444B, Paragraph 4.3.2.2.

4.5.12 X-ray examination of ballistic samples.-The cartridge prior to forwarding to the proving ground, shall be subjected to x-ray examination using x-ray equipment approved by the Government or any other method satisfactory to the Contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.-Preservation and packaging shall be in accordance with dwg. 8835105.

5.2 Packing.

5.2.1 Level A-.-The cartridge shall be packed in accordance with dwg. 8835104.

5.3 Marking.-Marking shall be in accordance with dwgs. 8835105 and 8835104 and Code of Federal Regulations, Title 49, Parts 100-199.

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5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use.-This specification covers the loading, assembling and packing for Cartridge, 40MM, Practice, M407A2.

6.2 Ordering Data.-See MIL-A-48078 (PA).

6.3 Submission of Inspection Equipment Desgins for Approval.-See MIL-A-48078 (PA). Submit equipment designs as required to Commander, Picatinny Arsenal, Attn: SARPA-QA-T, Dover, New Jersey 07801.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, Picatinny Arsenal, Attn: SAPPA-QA-A-S.

6.5 Submission of test data.-In addition to the normal distribution of records, when the cartridges are procured by the U.S. Army Armament Command one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.6 Cost of check test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

6.7 Standard deviation shall be calculated from the following formula.-

$$S = \sqrt{\frac{\Sigma(X-\bar{X})^2}{n-1}}$$

where: X= each individual value

$\bar{X}$= sample arithmetic mean $\frac{\Sigma X}{n}$

n= sample size

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$(X - \bar{X})$ = the sum of the squares of the differences between the sample mean and each individual value.

6.7.1 For a faster and easier method of arriving at the standard deviation formula, the following alternate method may be used:

$$S = \sqrt{\frac{n\Sigma X^2 - (\Sigma X)^2}{n \quad (n-1)}}$$

6.8 Premature burst.-A premature burst is considered to have occurred if the round functions in the gun bore or in flight.

6.9 Combining of proving ground tests.-When the contractor for the cartridge is also the contractor for one or more of the components thereof, the proving ground tests of the contractor may be combined with the proving ground tests of the cartridge, to save expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the additional tests specified in the component specification shall be conducted.

6.10 If any cartridge case, prior to insertion of the base plug and primer is suspected of containing an underweight or overweight propellant charge, based on visual inspection it will be removed from the lot and weighed on suitable scales. Any cartridge case found with less than 75 percent or more than 125 percent of the assessed propellant load will be classified as a critical defect and removed from the lot. Any cartridge case found to have a propellant weight out of drawing tolerance but within ± 25 percent of the assessed propellant load will be classified as a major defect and removed from the lot.

6.11 Proving ground test summary:

Test	Sample Size	Requirements
Functioning	First 3 lots - 144	See 3.4 and 4.4.3.9.1
	REG PROD 72	See 3.4 and 4.4.3.10.1

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1310-A037

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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